

42.510 Statistical Learning for Data Science

Unsupervised learning – clustering, recommendation systems

Bikramjit Das

Week 8, Lecture 1

Outline

- Recommendation systems
- Netflix prize
- Clustering
 - ▶ Hierarchical clustering
 - ▶ K-means clustering

Recommendation systems

- Personalize the user experience for online applications.
- Leverage data on items and customers (e.g., likes, purchase history).
- A key challenge: they must be fast and accurate.
- Common underlying analytics: clustering, collaborative and content filtering.

Netflix Cinematch

Cinematch

- Launched in 2000.
- Accurate within half a star 75% of the time.
- Need for a more advanced recommendation system.

Netflix prize

Key aspects of the competition

- 1 million USD prize for improving Cinematch by more than 10% on the test dataset.
- Training dataset: 100,480,507 ratings that 480,189 users gave to $\sim 18,000$ movies.
- Test dataset: 2,817,131 data points, used for the public and private leaderboard.
- Test predictions were scored against the true ratings (performance measured with the RMSE).

Netflix prize

Timeline

- October 2006: the competition was launched.
- November 2007: *BellKor* wins the 50,000 USD progress prize with a 8.43% improvement over Cinematch.
- June 2009: Team *BellKor Pragmatic Chaos* achieves 10.05% improvement.
- September 2009: Team *BellKor Pragmatic Chaos* officially wins the competition.

A few words from the winner: *From the Labs: Winning the Netflix Prize*

Clustering

Goal: to partition observations into distinct groups so that the observations within each group are quite similar to each other, while observations in different groups are quite different from each other.

Note: This is an *unsupervised problem*, because we are trying to discover clusters (or, in general, a structure) within a dataset (and not to predict the outcome of a given variable).

Clustering

Difference w.r.t. PCA

They both try to simplify the data, but:

- PCA looks to find a low-dimensional representation of the observations that explain a good fraction of the variance.
- Clustering looks to find homogeneous subgroups among the observations.

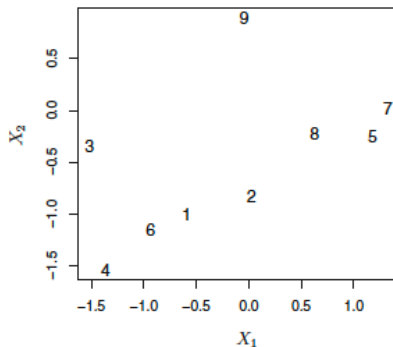
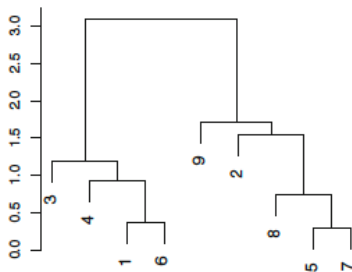
Clustering

Common methods

- *Hierarchical clustering*: we do not know in advance how many clusters we want (bottom-up, or agglomerative).
- *K-means clustering*: we partition the observations into a pre-specified number of clusters (top-down).

Hierarchical clustering

A motivating example (from James et al. (2023), Chapter 10.3):



Hierarchical clustering

Algorithm:

- ➊ Begin with n observations and a measure (such as Euclidean distance) of all the $n(n-1)/2$ pairwise dissimilarities. Treat each observation as its own cluster.
- ➋ For $i = n, n-1, \dots, 2$:
 - Examine all pairwise inter-cluster dissimilarities among the i clusters and identify the pair of clusters that are least dissimilar (that is, most similar). Fuse these two clusters.
 - Compute the new pairwise inter-cluster dissimilarities among the $i-1$ remaining clusters.

Hierarchical clustering

How do we define the dissimilarity between two clusters when we have multiple obs. in a cluster? Common types of **linkage**:

- *Complete* (maximal intercluster dissimilarity)
- *Single* (minimal intercluster dissimilarity)
- *Average* (mean intercluster dissimilarity)
- *Centroid* (dissimilarity between centroids)

K-means clustering

Notation: Let $C_1, \dots, C_K$ denote sets containing the indices of the observations in each cluster. These sets satisfy two properties:

- ① $C_1 \cup C_2 \cup \dots \cup C_K = \{1, \dots, n\}$
(each observation belongs to at least one of the K clusters)
- ② $C_k \cap C_{k'} = \emptyset$ for all $k \neq k'$
(clusters are non-overlapping)

K-means clustering

Idea: a good clustering is one for which the *within-cluster variation* is as small as possible. That means we want to solve the following problem:

$$\min_{C_1, \dots, C_K} \sum_{k=1}^K W(C_k)$$

We quantify the within-cluster variation with the *squared Euclidean distance*, that is:

$$W(C_k) = \frac{1}{|C_k|} \sum_{i, i' \in C_k} \sum_{j=1}^p (x_{ij} - x_{i'j})^2.$$

K-means clustering

Idea (cont'd): Putting the previous equations together, we get

$$\min_{C_1, \dots, C_K} \sum_{k=1}^K \frac{1}{|C_k|} \sum_{i, i' \in C_k} \sum_{j=1}^p (x_{ij} - x_{i'j})^2.$$

This is the optimization problem we want to solve. The K-means algorithm uses an heuristic search process to solve it.

K-means clustering

Algorithm:

- ➊ Randomly assign a number, from 1 to K , to each of the observations (initial cluster assignment).
- ➋ Iterate until the cluster assignment does not change:
 - For each of the K clusters, compute the cluster *centroid* (for the k -th cluster, the centroid is the vector of the p feature means for the observations in the k -th cluster).
 - Assign each observation to the cluster whose centroid is closest.

K-means clustering

Illustration of the search process for a toy case

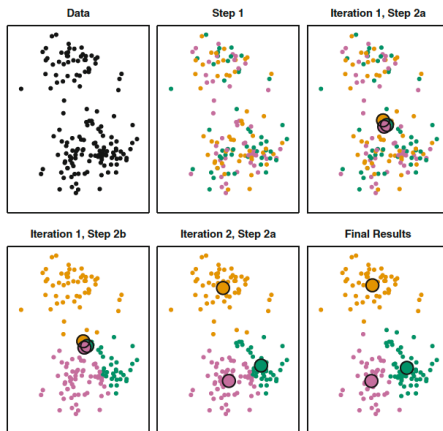


Figure 1: Source: James et al. (2023)

K-means clustering

A couple of considerations:

- The results will depend on the initialization.
- So, it is good practice to run the algorithm multiple times (with different initialization).
- We pick the configuration that minimizes the value of the objective function.

Final remarks

Some important decisions we generally have to make:

- Should the observations or features first be standardized in some way?
- For hierarchical clustering:
 - ▶ What type of linkage should be used?
 - ▶ Where should we cut the dendrogram in order to obtain clusters?
 - ▶ (What dissimilarity measure should be used?)
- For K-means clustering:
 - ▶ How many clusters should we use?

Clustering and Recommendation systems

- James et al. (2023) *An Introduction to Statistical Learning with Applications in R*, Chapter 10.3.

Netflix

- Koren Y. “The Bellkor solution to the Netflix grand prize.” Netflix prize documentation 81.2009 (2009): 1-10.
- *From the Labs: Winning the Netflix Prize*
- *Why Netflix's Algorithm Is So Binge-Worthy*
- *Why your Netflix thumbnails don't look like mine*